

BIG DATA COMPUTING

ID's last digit: 0 – 4

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OUTLINE

- ① Big Data Phenomenon
- ② Computational Challenges
- ③ Organization of the Course
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Big Data Phenomenon

Data

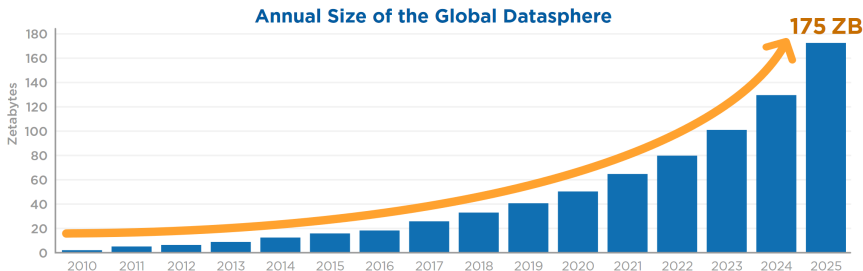
~~“Space is big. Really big”~~

(Douglas Adams, *The Hitchhiker's Guide to the Galaxy*)

Why is DATA growing so much?

- Technological progress:
 - Growth of storage capacity
 - Growth of communication bandwidth
 - Growth of computing capacity
- Reduction of ICT costs
- Pervasiveness of digital technologies: scientific research, health, business, politics, social interactions, ...

Big Data Phenomenon



From: *The Digitization of the World* (IDC, 2018)

How big is 175ZB?:

- 1 ZettaByte (ZB) = 1 trillion GB = 10^{12} GB;
- 175 ZB $\equiv$ 23 parallel stacks of DVD from Earth to Moon;
- Downloading 175 ZB at 1Gb/s takes > 43 million years

Big Data Phenomenon

The world continuously collects huge amounts of:

- SCIENCE** {
 - **Physical data**: from sensors, telescopes, particle physics experiments.
 - **Biological/medical data**: from genetic studies, patient monitoring, epidemic evolution analyses.
- Social Interactions** • **Human activity data**: from social networks, mobile devices, internet/web traffic, IoT systems. ↩
- BUSINESS** • **Business data**: from online stores, customer profiling, bank/credit-card/financial services, quality-of-service monitoring.

Big Data Phenomenon

The term **Big Data** relates to **two distinct issues**:

- **ISSUE 1:**

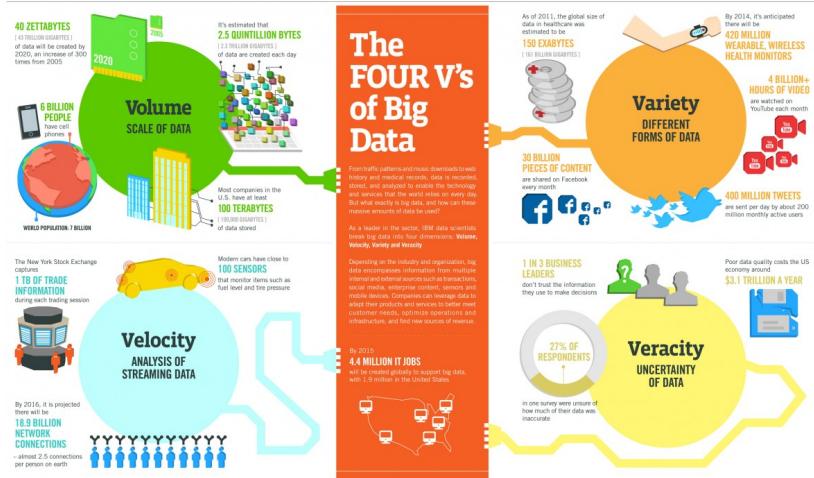
- Data produced everywhere;
- Need for automated analytics (vs human inspection);
- **Challenges:** identification of suitable analysis tools, data selection/preparation.

- **ISSUE 2:**

- Massive datasets need to be processed;
- Traditional (algorithmic) approaches are unsuited;
- **Challenges:** development of novel computing frameworks, novel solutions

This course focuses on ISSUE 2!

Computing Challenges



Sources: McKinsey Global Institute, Twitter, Cisco, Gartner, EMC, SAS, IBM, HP/DETC, SAS

IBM

Source: IBM Big Data & Analytics Hub

Computational Challenges

- **Volume:** processing huge datasets poses several challenges and requires a **data-centric perspective**.
- **Veracity:** large datasets coming from real-world applications are likely to contain *noisy, uncertain data*, hence **accuracy of solutions** must be reconsidered. *e.g data from telescopes*
- **Velocity:** sometimes, the data arrive at such a high rate that they cannot be stored and processed offline. Hence **stream processing** is needed.
- **Variety:** large datasets arise in *very different scenarios*. More effective processing is achieved by **adapting to the actual characteristics of data**.

The above issues require a

paradigm shift w.r.t. traditional computing.

Computational Challenges

To tackle the above challenges effectively, one needs:

- Platforms with:
 - High storage capacity and computing power
⇒ parallel/distributed architectures
 - Moderate costs
 - Ease of programming and management
- Focus on accuracy-resource tradeoffs, to cope with size, noise, and uncertainty of data
- Data-centric view
- Data stream processing (sometimes)

Big Data Computing Course

What will we learn?

- 1 **Novel computing/programming frameworks** for big data processing: theory and practice
- 2 **Key techniques** to process large-scale data
 - Rigorous setting (provable guarantees)
 - Application to fundamental data analysis primitives

Specific topics

- 1 **Distributed Frameworks**: MapReduce, Apache Spark (*partitioning; resource-accuracy tradeoffs*).
- 2 **Reducing input size**: clustering (*coreset approach, sampling*).
- 3 **Reducing output size**: frequent itemsets (*redundancy reduction, top-k approach*).
- 4 **Streaming Framework**: moments estimation (*reservoir sampling, probabilistic counting, sketches*).

Organization of the Course

Subdivision into classes

The students are subdivided into **two parallel classes** based on their ID's last digit (*same syllabus, homeworks, and exams*)

- Class 1 (prof. Pietracaprina): **last digit 0-4**
- Class 2 (prof. Silvestri): **last digit 5-9**

Lectures

Lectures will be online via Zoom. For Class 1 **meeting ID** is

<https://unipd.zoom.us/j/82862106641>

- Lectures will be recorded and uploaded to Moodle.
- For each topic, partial slide sets are made available in advance. Final versions (together with solutions to exercises) are uploaded after the topic is fully covered.
- *Attendance and active participation are strongly encouraged.*

Organization of the Course

Exam

- Final written exam (25 points) *taken individually*
- Homeworks: programming assignments (7+1 points)
 - Groups of 2-3 students (even from different classes)
 - 3 homeworks, approx. one every 3 weeks.
 - Use of Apache Spark on individual PCs (Homeworks 1-2) and on CloudVeneto (Homework 3)
 - All group members receive the same grade. The extra point is given if all homeworks submitted by the respective deadlines.

- *Oral exam at teacher's discretion*

*Cloud Infrastructure @ UNIBD
We'll use a cluster setup on Cloud Veneto*

Organization of the Course

Required background

- Java (preferred) or Python programming
- Basic algorithmics: asymptotic, worst-case analysis; sorting; fundamental algorithms and data structures; (e.g., lists, queues, stacks, hash tables, maps/dictionaries)
- Basic math tools, combinatorics, and probability.

Administrative Issues

Online tools

- **Course Moodle:**

<https://elearning.dei.unipd.it/course/view.php?id=6997>

- Announcements and student forum.
 - Infos: Zoom, contacts, textbooks, exam rules and sessions.
 - Material: slides, videos, exercises, articles.
 - Preliminary exams grades.
- **Uniweb:** Official exam lists and final grades.
 - **Exam Moodle (only one for the two classes):**

<https://esami.elearning.unipd.it/course/view.php?id=2242>

- *Formation of groups*
- Submission of Homeworks.
- Written tests (if online).

Administrative Issues

Contacts and office hours

- Teacher (prof. Andrea Pietracaprina):
andrea.pietracaprina@unipd.it
- TAs (dott. Diego Santoro and dott. Ilie Sarpe):
bdc-course@dei.unipd.it

Office hours (via Zoom) are by appointment (via Email). Teaching assistants should be contacted only for questions related to homeworks.

TODO: As soon as possible

- Each student must **register in the Course Moodle** (no password)
- Each student must **register in the Exam Moodle** (passwd **bdataEX**)
- Students must **form groups of size at most 3** for the homeworks. Once a group is formed, it must be registered in the Exam Moodle using the **Group registration** link.